



More equitable preconception health: paternal life course opportunities for better pregnancy, child, and family outcomes

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Men and partners are important contributors to the health of future generations, yet their own preconception health and wellbeing remain secondary considerations in research, practice, and policy. Siloed research has exacerbated this deficit. Clinical research typically has a narrow focus on proximal behavioural factors related to periconceptional events (eg, paternal dietary influences on the sperm epigenome), with social research focusing largely on postnatal parenting. Here, we update and reappraise the evidence for men's role in preconception health through a transdisciplinary review. Across biological and behavioural research, young men's early life course experiences have been shown to shape their own and their partner's preconception physical, emotional, and behavioural health. Moreover, focusing on men's preconception health offers a corrective for legacies of sexism, which place responsibility for intergenerational health solely on the birthing parent, and of racism and colonialism, which have disproportionately disrupted the familial and societal roles of Black and Brown men. We provide three case studies illustrating these ethical concerns and conclude that greater attention to young men would lead to more equitable and holistic preconception health interventions and policy.

Introduction

Nurturing preconception health is now widely recognised as being key to securing the health and wellbeing of future generations. Men and partners have an important role, yet their own preconception health and influences on maternal periconception health remain secondary considerations in research, practice, and policy. One driver for this disparity has been a historically narrow focus on the proximal clinical factors related to periconceptional events,^{1,2} such as paternal dietary influences on the germ line.² However, promotion of preconception health has evolved into a broader mandate encompassing upstream social, behavioural, and structural determinants of preconception health in people of reproductive age.³ Notably, Schoenaker et al's multi-national survey identified that both women and men consistently rank mental health, social, and financial support as high priorities before conception. Life course influences and healthy, engaged partners and fathers have been proposed as being key to maternal mental health, antenatal care attendance, and human potential.^{4,5} Nonetheless, the prevailing literature remains generally siloed, leaving large underexplored intervention spaces in the life course of boys and young men. Social science approaches largely focus on the transmission of culture or behaviours from parents to children, and clinical and biomedical approaches favour narrow mechanisms of intergenerational transmission (eg, via sperm epigenomics). These divisions leave wide gaps in understanding intergenerational biological phenomena and how they are expressed within and across populations and cultures, hindering translation to effective preconception interventions. Failing to consider the health and wellbeing of young men and fathers specifically exacerbates and perpetuates racial

and ethnic inequities in preconception, antenatal, and child health.^{6,7} Moreover, siloed research neglects insights from Indigenous, Asian, African, and other global perspectives^{7,8} that naturally de-centre adult individuals in favour of community-driven, intergenerational, and life course influences on preconception health.

This Review presents a holistic view of the contributions of men to preconception health, focusing on three aspects. First, we provide a transdisciplinary review of paternal life course effects on male and female preconception health, pregnancy, and offspring outcomes, highlighting underexplored targets for intervention. Second, we

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Key messages

- Boys and young men are an important but persistently underappreciated population for preconception interventions to improve intergenerational health
- Social and biological factors over the male life course influence male health and wellbeing at reproductive ages, and have important influences on maternal preconception health
- Supporting male preconception health is integral to health equity and reproductive justice, expanding the scope of interventions towards shared responsibility for parenthood
- Greater attention to the life course health and wellbeing of boys and young men is an important part in addressing intergenerational disparities arising from legacies of colonialism and racism
- Recognition of the broader implications of male health and behaviours across the life course on male and female preconception health is informing new targets for intervention and monitoring of preconception health indicators
- Greater appreciation for preconception social and biological roles of males calls for policy makers, public health agencies, clinicians, and men themselves to prioritise the roles and responsibilities of men in a more equitable vision to improve pregnancy, child, and family outcomes

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discuss equity implications of the exclusion of men in preconception and intergenerational health. Finally, we focus on the implications of this exclusion on future interventions and policy in paternal preconception health.

We define paternal preconception health as medical, behavioural, and psychosocial factors and their wider determinants in the life course of males that can influence pregnancy and birth outcomes and the health and development of children, not limited to the biological relationship to the pregnant person or developing fetus (panel 1). This integrated framework spans: (1) the early life and intergenerational social, political, and cultural determinants of health and behaviours of boys; (2) the biological role of fathers in healthy conception; and (3) the social role of biological fathers and potentially non-biological partners in protecting or harming the health of the pregnant person and child (figure 1).

Panel 1: Men and intersecting, sexual, gender, and family-role identities

A broad spectrum of sexual, gender, and partnership identities are described within this Review. We intentionally use common, but socially and contextually constructed, terms for males across the life course (ie, boys, men, and fathers). We recognise and allow for the conflation of gender, sexual, and social roles or their common intersections as they might present in various cultures. In this way, we are concerned not only with individuals who can produce viable male gametes post-puberty, but also anyone who would identify with these roles. This construction leads us to a broader definition of preconception health that encompasses acts proximal to oocyte fertilisation, and the totality of primordial social and behavioural factors commonly attributed to males that contribute to healthy conception, pregnancy, and offspring health and wellbeing.

Life course and intergenerational determinants of male preconception health

Search strategy and selection criteria

We conducted a semi-structured meta-review of recent literature (primarily from 2000 to 2025; appendix) and used three case studies as examples highlighting the broad-ranging ethical consequences of research and policy agendas that neglect the role of men and fathers. A holistic approach to preconception health that integrates sociobehavioural causes with biomedical mechanisms requires engagement with the specific contexts in which men and families live. Therefore, these case studies are not intended to represent the breadth of experiences of—or potential effects on—men from communities globally; rather, they illustrate the effects of two broader phenomena, colonialism and family-disrupting policies, on men's preconception health within specific contexts. Other communities and contexts will experience substantially different implications that can be elucidated through deeper anthropological inquiry.^{9,10} We conclude with a discussion of implications, including how these concepts contribute to overcoming barriers in acceptance and uptake of preconception intervention and policy.

Periconceptional determinants of sperm health: animal experimental models and human trials

We begin our review with advances in sperm health research that featured in the 2018 *Lancet* Series on preconception health.¹² By 2018, there was already substantial evidence from animal experimental models for the effects of paternal nutrition on sperm epigenomic programming and offspring outcomes.² Increasing experimental evidence has since documented the importance of varied paternal preconception exposures and related epigenomic pathways, including weight status and exercise,¹¹ environmental toxicants,^{12–16} and psychological trauma.^{17,18} However, this experimental evidence remains mixed as to whether interventions, such

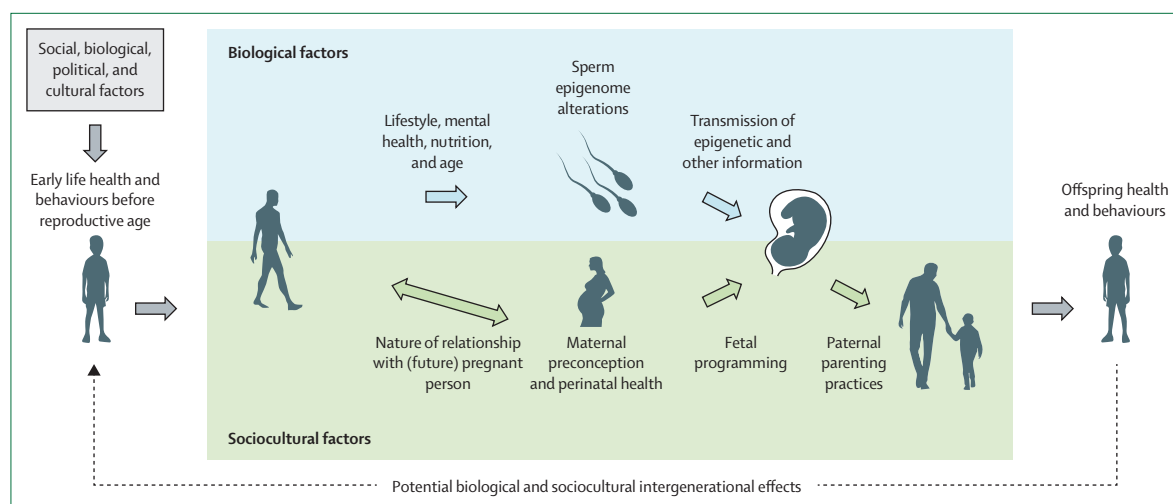


Figure 1: An integrated social and biological framework for male preconception health

Panel 2: Addressing challenges in interpreting findings in observational studies of paternal factors

Threats to inference

- Infrequent intent to recruit, or challenges in securing participation of, males in population-based studies³⁹
- Studies of infertility beset with confounding and limited generalisability³⁹
- Partitioning genomic or shared environment effects (or both), such as postnatal parenting⁴⁰
- Confounding by, or mediation through, correlated maternal exposures⁴¹
- Poorly specified exposure assessment and exposure timing³⁹

Potential solutions

- Application of target trial emulation methods, in which inclusion criteria and exposure conditions and timing are

clearly defined, could help to pinpoint and estimate realistic effects of paternal preconception factors⁴²

Approaches to account for maternal contributions

- Stratification: in mothers without overweight, paternal preconception BMI was associated with the child's bodyweight trajectory from infancy through to adolescence, as well as adverse glucose tolerance (ie, reduced Matsuda Index and an increased Homeostatic Model Assessment of Insulin Resistance score; both $p < 0.0001$)²⁵
- Adjustment: when adjusted for maternal folate intake, higher paternal preconception folate intake was associated with longer gestational periods (ie, 2.6-day longer gestation per 400 µg/day increase in folate, 95% CI 0.8–4.3)⁴⁴

as micronutrient supplementation,¹⁹ exercise,¹¹ or dietary changes,²⁰ can reverse sperm epigenomic damage. Although weight loss interventions (eg, bariatric surgery) show improvements in adult male cardiometabolic outcomes,²¹ the immediate benefits to fertility, pregnancy, and offspring outcomes are less clear.²²

Life course and intergenerational determinants of sperm health: longitudinal, non-experimental human studies

From a physiological perspective, preconception experiences of males were not believed to be able to affect offspring due to rapid sperm turnover. However, long-lasting epigenomic modifications occur throughout the male life course, including genome-wide reprogramming during primordial germ cell migration in early embryos. Furthermore, early life exposures, including nutrition, environmental toxicants, and other stressors, can affect pre-pubertal primordial germ cell differentiation to spermatogonia and thereafter to spermatozoa, and can influence post-fertilisation modifications via sperm histones, seminal fluid components,^{23,24} and signalling molecules (eg, small non-coding RNA, microRNA, tRNA-derived small RNA, and spermatozoa mitochondrial RNA).^{25,26} Consequently, these exposures over the entire male reproductive life course can produce long-lasting effects on offspring. Numerous reviews now support paternal pre-pubertal health influences on offspring development, including across generations due to intrauterine exposures.^{24,27} In a Swedish cohort study of 11561 men, paternal overnutrition (proxied by abundant harvests) during an important pre-pubertal period (ie, age 9–12 years) was associated with an increased diabetes mortality risk in sons (hazard ratio [HR] 1.84, 95% CI 1.21–2.79). Paternal grandfather exposure to abundant harvests pre-puberty was associated with a >3-fold risk of any cancer death and >4-fold risk of death from non-smoking-related cancers (HR 4.39, 2.02–9.53), with no corresponding associations for women, mothers, or

grandmothers.²⁸ These results highlight the importance of preconception sperm health in paternal early life and intergenerational health. More recent observational cohorts and consortia²⁹ are studying paternal effects on reproductive and pregnancy outcomes against a concern for global trends in declining sperm quantity and quality.³⁰ Mechanistic clinical studies also emphasise the transgenerational effects of paternal life course influences,²⁷ including paternal smoking and weight status;³¹ metabolic syndrome;³² mental health and psychological distress;¹⁸ alcohol, tobacco, and illicit substance use;³³ and advanced paternal age.^{34,35}

Male life course and intergenerational determinants under a holistic concept of preconception health

Although sperm health is one mechanism by which paternal preconception factors can influence offspring health, male health and social behaviours have broader influences, including through maternal biology and behaviours and paternal parenting (figure 1). Identifying both direct (ie, the sperm epigenome and paternal parenting) and indirect (eg, maternal health and behaviours) paternal influences can widen targets for preconception intervention. Paternal factors more commonly studied in mothers can have similar (or potentially larger) effects compared with their corresponding maternal factors; for example, in 529 090 Chinese couples, the odds of birth defects were 35% higher in infants born to fathers reporting any preconception alcohol consumption (adjusted odds ratio [OR] 1.35, 1.14–1.59).³⁶ Oldereid et al³⁷ found advanced paternal age at conception to be related to stillbirth, birth defects, autism spectrum disorders, and schizophrenia. Furthermore, a meta-analysis has associated paternal age greater than 45 years with autism spectrum disorder (adjusted OR 1.43, 1.33–1.53).³⁷ In more than 7 million father–child dyads, children of fathers with depressive disorders or subclinical symptoms were more likely to themselves have depression compared with

children of fathers without such symptoms (OR 1.42, 1.17–1.71).³⁸

However, numerous analytical challenges have hindered understanding of paternal preconception health effects (panel 2). Particular concerns include the disproportionate focus on maternal characteristics without addressing confounding by paternal factors and the use of highly selective, non-generalisable samples with paternal data.³⁹ For example, evidence for offspring epigenomic mediation of maternal periconception health effects is strongly attenuated when accounting for paternal confounding factors.^{41–43} Most studies do not distinguish between associations arising from preconception (eg, genomic or epigenomic) or postnatal (eg, shared environment or parenting behaviours) paternal influences, which could have opposing effects. For example, a systematic review found evidence for advanced paternal age being advantageous in certain offspring outcomes, such as educational attainment, externalising behaviours, and responsiveness to mental health treatments.⁴⁵ However, one twin study on the social development of offspring found that adverse consequences were associated with extremes in paternal age, with older fathers (ie, those aged >40–45 years) conveying preconception risk through genomic effects, and younger fathers (aged <20–25 years) conveying postnatal risk through non-genomic social factors.⁴⁶ In turn, determinants such as age at childbearing are influenced by prevailing policies, social norms, and cultural contexts that shape men's health directly and define the role of men in relationships.⁴⁷ The identification of relevant paternal preconception targets therefore requires more rigorous analytical methods and a more comprehensive view of drivers of paternal preconception health and behaviours. Social science research on paternal parenting and the role of fathers^{7,48,49} provides a guide for important social mechanisms, potential preconception health indicators,³ and targets for intervention.

Partner support of maternal preconception and pregnancy health

Partners are particularly important in shaping maternal health during periconception. Physical (ie, instrumental, practical, and material) and socioemotional (ie, supportive, neglectful, or abusive) relationships with partners are key drivers of women's preconception health. Partners have direct roles in decisions involving family planning, use of reproductive-related and general health care, and health-related behaviours at reproductive stages.⁴ Paternal support (eg, co-habitation or financial support) is associated with increased maternal participation in antenatal care,⁵⁰ reduced maternal alcohol and tobacco use,^{51,52} low birthweight reduction, and increased breastfeeding rates.^{50–54} Pregnant women reported healthier dietary choices when their partners were actively involved in cooking and grocery shopping, rather than when offering more passive support, such as giving

advice.⁵⁵ Women who perceived their partners as being more supportive had substantially lower levels of perinatal depression and anxiety,^{56–58} whereas avoidant partner coping styles are associated with higher rates of maternal depression.⁵⁹ Other meta-analyses support strong correlations between paternal and maternal depression across the perinatal period.^{60,61} Poor maternal mental health in the periconception period adversely influences cognitive and socioemotional outcomes in offspring,⁶² whereas positive maternal mental health beneficially impacts the child's language and social communication skills.⁶³

The life course development of executive functions (ie, essential cognitive processes that allow individuals to maintain cognitive control and adapt behaviour to challenging settings⁶⁴) is central to a partner's ability to be supportive and caring.⁶⁵ Adverse childhood experiences (ACEs), such as poverty, maltreatment, and neglect, are robustly related to executive function deficits throughout life.⁶⁶ Men with executive dysfunctions, such as reduced self-control, tend to show avoidant coping mechanisms, such as alcohol consumption in response to negative emotions.⁶⁵ These coping mechanisms can influence maternal alcohol consumption and other risk-taking behaviours,⁶⁷ and can lead to intimate partner violence.⁶⁸ Executive function deficits can raise tension between partners, increasing depressive and anxiety symptoms in pregnant women,^{69,70} and subsequent adverse pregnancy and developmental outcomes.^{67,71,72} Conversely, male partners who use effective coping strategies against adverse experiences are better equipped to provide a supportive family environment.⁷³

The male early life course and determinants of postnatal parenting and child development

Early life factors in men that influence offspring outcomes via paternal parenting are a key component of preconception health. Clinical factors that affect sperm and the offspring epigenome—such as paternal mental health, age, and physical health status preconception—can also influence relationship quality, coparenting dynamics, family conflicts, infant bonding, and infant neurodevelopment through behavioural pathways established before reaching reproduction and parenthood.^{29,74} Determinants of paternal parenting are particularly salient, as paternal parenting can be more important than maternal parenting for outcomes such as child weight status.⁷⁵ One core determinant is paternal mental health. Men's ACEs are widely recognised as important contributors to anxiety and depression in adolescence and young adulthood,⁷⁶ persist into reproductive age,⁷⁷ and are exacerbated during the transition to parenthood.^{78,79} Male ACEs are robustly related to their own⁸⁰ and their partners'⁷⁶ preconception anxiety and depression, which in turn adversely affect their ability to parent.⁷⁶ Boys' exposures to ACEs are common and have lasting consequences,

contributing to the 8–10% of fathers experiencing depression from preconception to postpartum.^{60,77–79,81} A 20-year follow-up of adult men found those who reported subclinical depression or anxiety in adolescence and age 20–29 years were almost five times more likely to report psychological distress (ie, a General Health Questionnaire-12 score ≥ 3) during their partner's pregnancy, compared with men without reported adolescent depression or anxiety.⁷⁷ The early origins of paternal distress, including through executive function development, could help explain why perinatal-focused interventions are largely ineffective in reducing paternal depression or its offspring sequelae.⁸² By contrast, efforts to mitigate the effects of parental childhood ACEs on preconception mental and physical health highlighted positive relationships and experiences, including “perceiving support, protection, trust, and belonging from family, friends, and the community”.⁸³ Development and maintenance of such positive lasting relationships in boys and young men could be key to improving immediate preconception health and preventing ACEs in the next generation,⁸⁴ further emphasising the need to better understand the role of fathers within context, community, and family structures.⁸⁵

Neurobiological and behavioural changes in expectant fathers

Support for intergenerational transmission of paternal caring is evident in recent hormonal and neuroimaging research. Like women, men also experience hormonal changes over the course of pregnancy, including reductions in testosterone, vasopressin, and oestradiol over the postpartum transition.^{86,87} These changes are associated with increased parenting sensitivity and engagement in subsequent years. A meta-analysis of 50 studies (N=7080 men)⁸⁸ found moderate effects of lower testosterone on parenting quality. In Filipino men,⁸⁹ fathers showed greater decreases in salivary testosterone than childless men, and those who grew up with engaged, supportive fathers had even lower salivary testosterone.⁹⁰ Preliminary research suggests microstructure and macrostructure brain changes in expectant and new fathers,⁹¹ mirroring well characterised changes in maternal brain structure.^{92,93} In small, cross-national studies,⁹¹ transition to fatherhood was associated with context-specific reductions in cortical volumes of the default mode network, which is associated with the resting state, internal narrative, planning, and visual networks. In turn, greater reductions in these cortical volumes, which could be related to identifying and responding to infant needs, have been associated with self-reported increased engagement and bonding with infants.⁹⁴ Although understudied, these mechanisms support a physiological basis for life course effects on paternal preconception and offspring-related postnatal behaviours centred on engagement and caring.

Equity implications of neglecting the role of men in preconception health

Case 1: pre-empting maternal blame and responsibility

A major consequence of medicalisation and pathologising pregnancy is the potential for women and childbearing people to become targets of blame and the sole bearers of responsibility for offspring outcomes.^{95–97} Acknowledging social and cultural biases is important as they influence not only the types of research conducted, but potentially impede opportunities for developing, evaluating, and translating male preconception interventions and policies. The consequence of such biases is particularly acute when there is evidence that paternal preconception effects can be stronger than maternal factors. In ASD, for example, paternal age can have an even stronger effect than maternal age; in meta-analyses, each 10-year decrease in maternal age was associated with a small and statistically non-significant 7% reduction in the odds of autism (OR 0.93, 0.69–1.24), compared with a 26% reduction for lower paternal age (0.74, 0.64–0.86; one-sided interaction p value 0.09).⁹⁸ Although understudied, romantic partners and spouses have important roles in fertility decisions, health behaviours, and socioemotional and instrumental support. Solely focusing on optimising maternal preconception health therefore leaves a gap in men's understanding of, and preparedness for, preconception health.⁸¹ A growing expectation of more equal participation in family planning, pregnancy, and childcare⁷⁶ has not been met with a commensurate increase in efficacious interventions. The absence of male-focused or partner-focused materials is often cited as the major barrier to men participating in antenatal care,⁹⁹ as reflected in clear gaps¹⁰⁰ in recommendations for, and provision of, men's preconception health services. A US study suggested two-thirds of men had unmet preconception health-care needs,¹⁰¹ and most preconception care guidelines internationally omit guidance for men.¹⁰² Paternal

Panel 3: Paternal–maternal relationship dynamics as distinct preconception health indicators

- Most studies on socioeconomic predictors of preconception and antenatal health focus solely on maternal or aggregate, household-level factors¹⁰⁴
- However, social mechanisms tend to drive convergent health-related and risk-taking behaviours within partnerships; socioeconomic (eg, educational) gradients in poor health are therefore underestimated when women are considered in isolation from partners¹⁰⁵
- Directionality matters: when examined in sequence, paternal perinatal depression predicted worsening symptoms in postpartum mothers, but not the reverse¹⁰⁶
- These findings suggest the importance of considering interactions between partner characteristics as distinct indicators of preconception health beyond those characteristics in isolation

preconception care has almost exclusively focused on proximal, individualised, and clinical determinants of healthy conception (eg, smoking cessation),¹⁰³ neglecting the role of early life described in this Review. Finally, treating partners as individuals leaves large gaps in developing important preconception health indicators and potential interventions that are related to couples' dynamics above and beyond their individual actions (panel 3). A holistic view of men's roles in preconception health would help shift the focus away from mothers as the sole bearers of responsibility for offspring health.

Case 2: disproportionate impacts of colonialism on the health of young native and Indigenous men

Viewing men as individual actors, separate from family, ancestry, and responsibility for raising the next generation, is at odds with many traditional Indigenous, Asian, African, and other cultures and communities. Research across diverse nations and settings often shows fathers to be nurturing, highly engaged in childcare, and situated within expanded family structures supporting child development, without the typical role delineations found in the dominant cultures of modern, post-colonial societies.^{8,107–110} For example, these nurturing, pro-social roles were seen as aspirational goals in middle-age and older-age African American men's concepts of manhood.¹¹¹ Among Indigenous people whose connections to land, traditions, and cultural practices have been disrupted and erased by colonialism, life course health consequences are disproportionately borne by young men who experience high burdens of ACEs.¹¹² These men are also disconnected from the traditional roles and practices that facilitate community, support, kinship, and the sense of belonging that buffers against adversity.⁸³

Among Native Hawaiian boys and young men, the 1893 overthrow of the Sovereign of the Kingdom of Hawai'i Queen Lili'uokalani, and subsequent annexation of Hawai'i by the USA, has led to widespread actions that affect preconception health inequities.¹¹³ The seizure of lands and political control resulted in cultural loss, discriminatory policies and practices (eg, banning of native language), environmental hazards, and persistent economic and social disadvantages. In turn, this loss of self-determination led to historical trauma (ie, inter-generationally transmitted, cumulative, and collective emotional harm) that have manifested as ACEs with attendant sequelae, including obesity, substance use, stress, and depression. Native Hawaiian 'ōpio (youth) have associated historical loss of land and traditional practices with feelings of pain and sadness, but also with anger, loss of control, self-harm and physical violence, hopelessness, and avoidant behaviour.¹¹⁴ The link between historical trauma and executive function, therefore, could underlie poorer educational outcomes and higher rates of suicide, arrest, and incarceration in young Native Hawaiian men.^{8,115} These men might also be more affected by cultural loss than young women due to differential

impacts of changes in gender-attributed societal roles.¹¹⁶ Traditionally, Native Hawaiian men had roles as warriors, navigators, farmers, fishermen, builders, carvers, and healers. These men were responsible for the health and wellbeing of their 'ohana (family) and kaiāulu (community), and had important roles in child rearing, especially in supporting younger males to become healthy and productive adults.¹¹⁷ Around age 8 years, a boy would leave his mother's side to join the *Halemuia* men's house to learn his occupation, familial and societal obligations, and the associated socioreligious practices from other men. However, colonisation, including the introduction of Christianity, replaced or banned such practices in favour of labour, parenting, and child-rearing practices that predominated in the Anglo-Eurocentric continental USA. The loss of these roles within communities came with racist judgments of inferiority and the loss of social and communal ties that buffer against adversities. This loss of cultural practices harms preconception physical and mental health, impairing men's ability to fully participate as supportive partners and fathers. Moreover, over-policing and loss of generational wealth, livelihoods, and supportive communities all contribute to disproportionate incarceration, precluding the physical presence needed to support partners before, during, and after pregnancy. Conversely, the reclamation of Native Hawaiian male cultural practices and fellowship could serve to undo these harms and foster improved preconception health (panel 4).¹⁰⁹

This case study is entirely consonant with our conceptualisation of the role of men in holistic preconception health. A renewed focus on men (beyond mere providers of sperm) as members of partnerships, families, and communities, and on the importance of community caring to buffer against early life adversities, is at the core of male preconception health.

Case 3: harmful social policies that exacerbate racial and socioeconomic perinatal and life course disparities

The effects of social policies on paternal engagement are profound. Numerous policies either discourage or outright restrict fathers' involvement in the preconception and early life stages of their children. For instance, although paternity leave is lauded to increase fathers' involvement in child engagement and caregiving, meagre provisions and protections continue to signal their lesser importance. Moreover, social policies in some countries (eg, the USA) are restrictive in ways that sideline paternal participation in childcare.^{6,119} Policies restricting government benefits (eg, child subsidies, nutritional benefits, and housing) to single parents—or to mothers specifically—disincentivise paternal engagement in antenatal care and child rearing, especially in formal capacities, and can even disincentivise marriage.¹²⁰ In the USA, these disincentives are particularly acute for Black fathers on low income, who can be driven away from their children and paternal responsibilities as a

Panel 4: Addressing adverse childhood experiences—the revitalisation of cultural practice

- Adverse childhood experiences (ACEs) are a key upstream cause of male preconception health disparities. Policies for addressing ACEs include: adopting national strategies, enhanced data and research, strengthening prevention and early intervention, integrating ACEs into educational policy, support for families, cross-sector collaborations, and public awareness campaigns.¹¹⁸
- More specific and practical models of these policies are needed for boys and young men. For example, the practice of Lua, the traditional fighting art of Hawai'i, could be used to address high rates of ACEs, interpersonal violence, substance use, and academic underachievement in Hawaiian males through early prevention and targeted efforts to engage them in building a sense of community, centred on cultural revitalisation.
- The Hawaiian Renaissance movement began in the 1970s to uplift native culture, including language, traditional dance (ie, hula), navigation, the protection of cultural and natural resources, and self-determination.
- Since the 1990s, attempts have been made to specifically uplift the health and wellbeing of Native Hawaiian males (ie, kāne), notably by revitalising interest in Lua as a traditional, male-oriented practice.
- Drawing from the Halemua concept of establishing a supportive community of males, from 'ōpio (youth) to kūpuna (elders), grassroots Halemua-like groups emerged to engage their male members in revitalising socioreligious ceremonies that had been banned or abandoned due to Christianisation. These groups sought to reconnect men to their Hawaiian culture, identity, and support systems.
- In 2006, 'Aha Kāne—the Foundation for the Advancement of Native Hawaiian Males—was established to improve the wellbeing of Native Hawaiian kāne, their families, and communities. 'Aha Kāne engaged men of all ages in cultural activities designed to reconnect them to their native language, ceremonies, and other practices, while instilling strong 'ohana and kaiāulu values and encouraging them to act as positive role models for the next generation of young men.

result of punitive child support policies.^{121,122} Barriers to paternal participation are exacerbated by decades of disproportionate imprisonment of minoritised young men, with adverse consequences of both their development and their ability to support mothers of their children.¹²³ Despite these substantial barriers, minoritised, non-resident fathers—especially Black Americans—often remain more engaged with children than non-resident White fathers, indicating a broader understanding of parenting beyond rigid, nuclear family-oriented marital arrangements.⁷ Family-disrupting policies are not exclusive to Black American men, and the global dynamics of migration due to economic policies, war and conflict, and other hazards affect men and fathers differently. However, little research exists on the implications for preconception and perinatal health.^{124,125} Continuing to investigate and address policies that impede or otherwise artificially shape men's participation¹²⁶ in families is crucial for supporting reproductive justice and creating a more equitable framework that supports the important role of fathers in preconception and early childhood development.

Implications for current and future preconception interventions and policy in boys, men, and partners

So far, we have reviewed how narrow views on the role of men on pregnancy and offspring health restricted the scope and opportunities for initiatives and interventions to improve paternal preconception health, such that these lag well behind those supporting women. Half of international clinical preconception care guidelines do not include recommendations for male preconception

health,¹⁰² and health services often fail to provide information for men.¹²⁷ Although most men do not receive preconception education and care,^{101,128,129} the case to do so grows more urgent: male obesity, smoking, mental illness, and medications that impair male reproductive health are common from a young age,^{128,130–132} and are becoming increasingly prevalent.^{130,131} Moreover, these factors are stratified by race and ethnicity,¹³⁰ underscoring the need for culturally sensitive paternal preconception health care.

Individual-level barriers

Raising awareness about paternal preconception health is essential. Research suggests that men generally wish to be involved in planning and preparing for pregnancy and parenthood once they understand how their health affects pregnancy and child outcomes.¹³³ Based on consistent findings from surveys and focus groups in the USA, Europe, and Africa, most men agree with statements such as “as a man I can contribute to a healthy pregnancy and child”.^{134–137} When asked how they can contribute, men often describe roles as “ensuring a healthy relationship with [their] partner” and “making sure [they] have a house and are financially stable”.³ However, men are often unaware of the full extent of their potential contribution, based on common misconceptions that “pregnancy and child outcomes are almost solely influenced by the woman's health” and “[they are] already healthy enough”.^{135,137} Research in sub-Saharan Africa indicated that men desired greater engagement in improving maternal and child health, but felt that their ability to do so was limited by culture-specified gender roles focused on providing for and advising their families.¹³⁸ Additional barriers, such as lower rates of

accessing health care compared with women and sparse knowledge about male fertility and preconception health among health-care professionals,^{138–140} further hinder improvements to male preconception health. More research is needed to establish the best ways to educate and support men, but existing studies consistently highlight an important role for health-care professionals, followed by trusted online resources.^{133,135,136,141}

Social and political barriers

In addition to social policies and personal factors (eg, health status, substance use, and mental health), multiple societal factors affect a father’s ability and opportunity to engage with their child.¹⁴² For example, traditional gender norms and cultural expectations often maintain the notion that child rearing is predominantly a maternal responsibility. Engaged fathers, particularly from minoritised communities, can be viewed with distrust¹⁴³ or otherwise excluded.¹⁴⁴ Economic factors, such as job insecurity and inadequate wages, can hinder fathers’ ability to participate fully in their children’s lives.¹⁴⁵ Overcoming these barriers is essential for aligning with broader reproductive justice goals. Culturally competent initiatives that promote mental health management, flexible work arrangements, and overall wellness can empower men and shift societal perceptions to encourage paternal engagement. Efforts that provide structural support tailored to the unique needs of fathers, especially those from minoritised communities, are crucial. Initiatives that address social determinants of health, such as access to health care, housing, and education, can enhance men’s preconception health and engagement as fathers.

Facilitators

Although men themselves are important facilitators for improving preconception health, shifting social norms and ensuring that health-care systems support men’s involvement in parenthood preparation is also essential. Some men express skepticism over their need for

immediate behavioural change,¹³⁵ but many articulate their desire to support healthy pregnancies and offspring, persevere in family engagement, and even recognise more direct contributions (eg, sperm health). Moreover, traditional cultures, when reinvigorated and not impeded by policies, are strong drivers of pro-social male engagement in families, partnerships, and child rearing. The immediate opportunity is for providers and systems to allow these strengths to flourish, alongside health-care provision for boys’ and young men’s holistic wellbeing.

Vision of male preconception health interventions

We present a vision for potential policies that embrace a broader understanding of the importance of the male life course and the effects of men on maternal preconception health (figure 2). Such an approach recognises that promoting the health and wellbeing of women and men before and during the first 1000 days after conception will have lasting biological effects on the male offspring that support their later preconception health as future fathers. ACEs underlie life course determinants of male preconception health and behaviours influencing maternal health and wellbeing. A transgenerational life course approach, starting with a focus on the first 1000 days of life, should be combined with societal interventions that prevent ACEs, alongside individual-level, culturally relevant remediation for those who experienced ACEs (panel 4). A roadmap has been proposed for a coordinated, preventative, and remediative response at all levels of policy making in the UK.¹¹⁸ Biomedical determinants of paternal preconception health, such as genetic and congenital conditions, disability, subfertility, and serious health conditions, often act from early childhood; consequently, their management in childhood should be protective of future reproductive health (eg, avoiding testicular irradiation for childhood cancer).¹⁴⁶ By promoting health literacy as a route to healthy behaviours, policy measures centred on the education system (ie, from the early years to higher education) can support paternal preconception health. A cluster

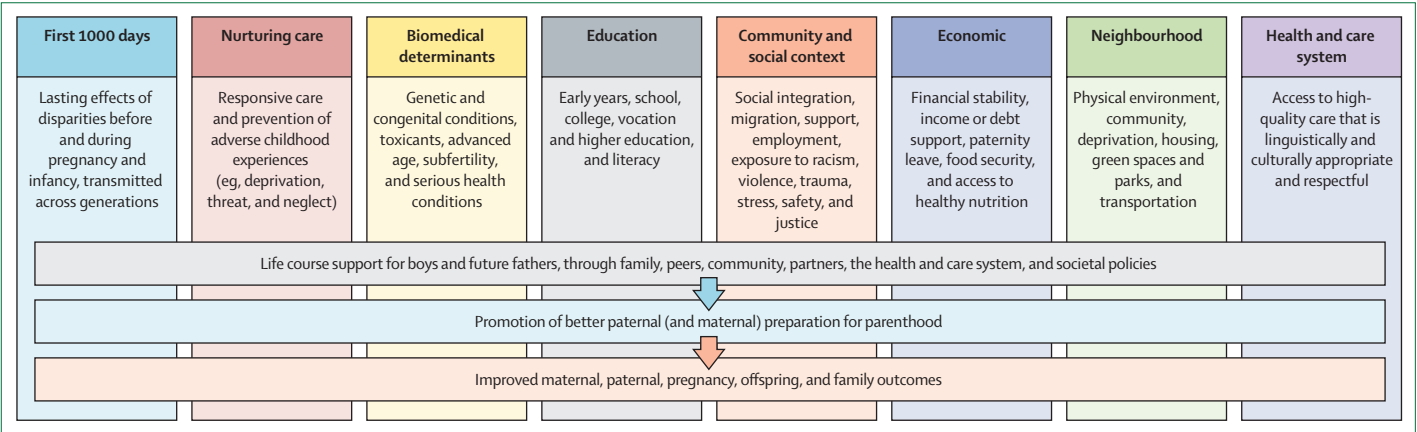


Figure 2: Holistic life course determinants of male preconception health

randomised trial showed that incorporating Southampton LifeLab's Me, My Health and My Children's Health engagement programme into the school science curriculum had lasting benefits on how critically adolescents considered their health-related behaviours, with similar effects in both boys and girls.¹⁴⁷

Public health policy makers now recognise the importance of wider determinants of health. Successful models have applied these determinants to women's preconception health,¹⁴⁸ but such an approach has yet to be embraced in policies affecting men. These policies should address the community and social context, and economic and fiscal measures, including reducing food insecurity and supporting positive neighbourhood and physical environments. Many opportunities exist for health-care systems to provide future fathers with access to high-quality, culturally appropriate preconception care (eg, medication reviews for men with long-term conditions such as epilepsy and mental ill health). The training of health-care providers to discuss preconception health with both men and women is an unmet need. Such provision should be combined with public information initiatives on how biosocial factors, such as lifestyle choices, advanced paternal age, and health conditions, can affect fertility and offspring health.

Conclusion

This Review, in elucidating the far-reaching role of men in preconception health, reinforces the need for holistic strategies beyond participation in preventative health care in early adulthood,¹⁰³ into care for the early development of boys and young men. Paternal preconception health necessitates engagement and participation from clinicians and education, social, and mental health-care professionals, along with traditional reproductive specialties. The disproportionate, intergenerational harms borne by Indigenous, non-White, and socioeconomically disadvantaged men and their families as a result of poor attention to young men's health must also be acknowledged. Such a holistic framework will also aid identification of how social forces that shape ideals of masculinity are linked to young men's mental health and their roles as caretakers, and thus serve as additional drivers of preconception health.¹⁴⁹ A transdisciplinary approach to preconception health research across the life course, integrating social and biological pathways, and explicitly including a more equitable consideration of men and partners, provides a strong opportunity to deliver a comprehensive preconception health strategy.

Contributors

JYH, FML, DS, and KMG contributed to conceptualisation and overall methodology. JYH and FML contributed to review of the literature and drafting original material for the entire manuscript. MZLK, KKS, and GSKC contributed to review of the literature and drafting portions of the Life course and intergenerational determinants of male preconception health section. LNH and JKK contributed to drafting portions of the

Equity implications of neglecting the role of men in preconception health section. DS and KMG contributed to drafting portions of the Implications for current and future preconception interventions and policy in boys, men, and partners section. JYH, FML, DS, and KMG contributed to the figures. JYH and KMG contributed to managing and administering the project. DS and KMG contributed to supervising the project. All authors contributed to substantive revisions and edits to the manuscript.

Declaration of interests

We declare no competing interests.

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